

University Semester Examination Solutions: Electrical Machines-II

(Part-II: 3-Phase Transformers & Tap Changers)

Course Instructor Reference: SR Sir
Electrical Engineering Department

2025

PART-II (Full Marks: 50)

Instructions: Answer any THREE questions. Question no. 3 carries 18 marks.

Note: As per your request, all questions and sub-parts are solved comprehensively below.

Question 1. Correct and/or justify following.

[4 × 4 = 16 Marks]

Problem Statement 1(i):

Delta-star is preferable than star-star without neutral three phase transformer.

Justification:

Thought Process:

1. Identify the fatal flaw in a 3-wire $Y-Y$ connection: The zero-sequence (third harmonic) components of the magnetizing current cannot flow because $i_a + i_b + i_c = 0$.
2. Without 3rd harmonic current, the core flux becomes flat-topped.
3. A flat-topped flux induces a highly peaked, distorted phase voltage which stresses the insulation.
4. A Delta (Δ) primary provides a closed circulation path for these in-phase 3rd harmonic currents, keeping the core flux strictly sinusoidal and the induced voltages safe.

This statement is **correct**. In any transformer, the non-linear magnetic properties of the core require a peaked magnetizing current (containing a strong 3rd harmonic) to produce a purely sinusoidal magnetic flux.

In a three-phase system, 3rd harmonic currents are co-phases (they all reach their peak at the exact same time in all three phases). In an ungrounded star-star ($Y-Y$) transformer, Kirchhoff's Current Law at the neutral dictates that $i_a + i_b + i_c = 0$. Therefore, these co-phases 3rd harmonic currents have no return path and are forced to zero.

Because the required 3rd harmonic current cannot flow, the resulting core magnetic flux $\phi(t)$ becomes flat-topped. According to Faraday's Law ($e = -N \frac{d\phi}{dt}$), taking the derivative of a flat-topped flux wave yields a highly peaked phase voltage containing large 3rd harmonic voltage components. This peaked voltage places severe dielectric stress on the transformer insulation and can lead to breakdown.

Conversely, a Delta-Star ($\Delta-Y$) configuration contains a closed delta primary loop. This closed loop acts as a local short-circuit path that allows the co-phases 3rd harmonic currents to circulate freely

within the primary winding. As a result, the core flux remains perfectly sinusoidal, and the secondary phase voltages remain undistorted. Hence, $\Delta - Y$ is strictly preferred over ungrounded $Y - Y$.

Problem Statement 1(ii):

The cross section of the central limb of a shell type transformer is usually designed to have larger cross section to avoid flux saturation.

Justification:

Thought Process:

1. Visualize a single-phase shell-type core. The windings are placed around the central limb.
2. The entire primary magnetic flux (Φ) is generated within this central limb.
3. Upon reaching the upper and lower yokes, this flux splits equally into two parallel return paths down the outer side limbs ($\Phi/2$ each).
4. To maintain a constant maximum flux density ($B = \Phi/A$) and prevent saturation in the center, the central limb must have twice the area of the side limbs.

This statement is **correct**. In a single-phase shell-type transformer, both the primary and secondary windings are concentrically wound around the central limb. Therefore, 100% of the total working magnetic flux (Φ) generated by the windings must pass through this central leg.

When the flux reaches the yokes (top and bottom of the core), the magnetic circuit divides into two symmetrical parallel return paths. Exactly half of the flux ($\Phi/2$) travels through the left outer limb, and the other half ($\Phi/2$) travels through the right outer limb.

To optimize the use of magnetic steel and ensure that the maximum flux density ($B_{\max}$) remains uniform throughout the entire core structure without causing localized magnetic saturation, the cross-sectional area must be proportional to the flux it carries:

$$B_{\max} = \frac{\Phi}{A_{\text{center}}} = \frac{\Phi/2}{A_{\text{outer}}}$$

$$\implies A_{\text{center}} = 2 \times A_{\text{outer}}$$

Thus, the central limb must be designed with twice the cross-sectional area of the outer limbs to safely accommodate the total flux and avoid saturation.

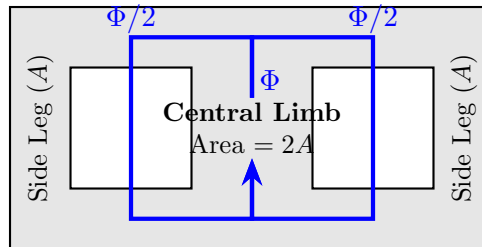


Figure 1: Flux division in a shell-type transformer dictating the central limb cross-section.

Problem Statement 1(iii):

The triplen order flux, sometimes links the tank of transformer and produces loss in the same.

Justification:

Thought Process:

1. Triplen harmonics ($3^{\text{rd}}, 9^{\text{th}}, 15^{\text{th}}$) are zero-sequence components, meaning they are completely in-phase across all three phases.
2. In a standard 3-limb core transformer, fluxes in the three legs try to flow in the same vertical direction simultaneously at any given instant.
3. Since there is no 4th return leg, the sum of these fluxes cannot return through the core ($\Phi_{3a} + \Phi_{3b} + \Phi_{3c} \neq 0$).
4. They are forced to exit the core and return through the air/oil and the solid steel walls of the transformer tank.
5. This alternating flux cutting through the solid steel tank induces severe eddy currents, causing massive I^2R heating losses in the tank.

This statement is **correct**. In a standard three-phase, 3-limb core type transformer, there are only three vertical iron legs. Under conditions where third-harmonic (or any triplen order) fluxes are present, a unique problem arises.

Triplen harmonics are zero-sequence, meaning the fluxes in phase A, phase B, and phase C are identically in phase with each other ($\Phi_{3a} = \Phi_{3b} = \Phi_{3c}$). At any given instant in time, all three triplen fluxes attempt to flow in the exact same direction (e.g., upwards) through all three core limbs simultaneously.

Because they are all flowing upward, they require a downward return path to complete the magnetic circuit. Since a 3-limb core does not have a fourth neutral return leg, these fluxes are forced out of the high-permeability iron yokes. They must jump into the surrounding medium (transformer oil) and find their return path through the easiest available magnetic route: the **mild steel walls of the transformer tank**.

As this high-frequency alternating triplen flux cuts through the solid, unlaminated steel of the transformer tank, it induces large localized electromotive forces (EMFs). These EMFs drive heavy circulating **eddy currents** within the tank walls. This results in significant localized I^2R heating (often called stray losses or tank heating), which lowers the efficiency of the transformer and can cause hazardous hot spots.

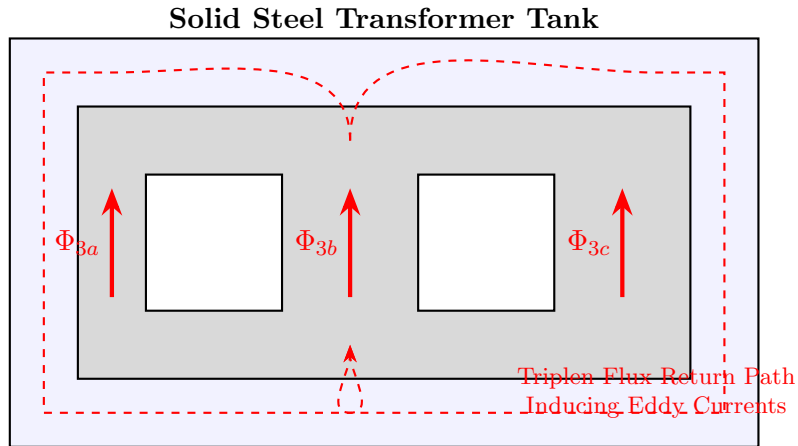


Figure 2: Triplen fluxes forced to return via the tank walls in a 3-limb core.

Problem Statement 1(iv):

Vector group formation is unavoidable with 30 degree phase shift in case of Delta-star or Star-delta transformer.

Justification:

Thought Process:

1. A phase voltage on the secondary must physically be in-phase with the phase voltage on the primary wound on the same limb.
2. In a Star (Y) winding, the line voltage is the vector difference of two phase voltages ($\vec{V}_{AB} = \vec{V}_{AN} - \vec{V}_{BN}$). This geometry intrinsically creates a 30° shift between line and phase.
3. In a Delta (Δ) winding, line voltage equals phase voltage.
4. Therefore, mapping a Delta line to a Star line inherently introduces a $\pm 30^\circ$ difference between primary and secondary line voltages.

This assertion is **correct**. In any transformer, the induced phase voltage in the secondary winding must be perfectly in phase with (or exactly 180° opposite to) the primary phase voltage on the same core limb, due to the shared magnetic flux.

However, the external terminals of the transformer represent **line voltages**.

- In a **Delta** (Δ) connection, the line voltage is identical to the phase voltage ($\vec{V}_L = \vec{V}_{ph}$).
- In a **Star** (Y) connection, the line voltage is derived from the phasor subtraction of two phase voltages (e.g., $\vec{V}_{AB} = \vec{V}_{AN} - \vec{V}_{BN}$).

From standard three-phase phasor geometry, subtracting two vectors that are 120° apart results in a resultant vector that is scaled by $\sqrt{3}$ and rotated by exactly 30° :

$$\vec{V}_{AB(Y)} = \sqrt{3}V_{ph} \angle \pm 30^\circ$$

Because a $\Delta - Y$ or $Y - \Delta$ transformer has one side experiencing this geometric 30° shift and the other side not, the line-to-line output voltage will always be intrinsically shifted by 30° (either leading or lagging) relative to the primary line-to-line input voltage. This unavoidable displacement gives rise to vector groups designated by clock numbers 1 (-30° lag) or 11 ($+30^\circ$ lead).

Question 2. Vector Groups and Parallel Operation**[9 + 7 = 16 Marks]****Problem Statement 2(i):**

Draw the connection diagram and phasor diagram of the following connections. Show how Dy1 and Yd11 transformers can be connected in parallel successfully.

Answer:**Thought Process:**

1. Part 1: Draw Dy1. Delta primary, Star secondary. The secondary line voltage lags the primary by 30° (1 o'clock).
2. Part 2: Draw Yd11. Star primary, Delta secondary. The secondary line voltage leads the primary by 30° (11 o'clock).
3. Part 3: Paralleling them. A Dy1 has a phase displacement of -30° . A Yd11 has a phase displacement of $+30^\circ$. A direct parallel connection is impossible (ΔV exists).
4. Solution: If we intentionally swap two of the input phases (e.g., B and C) feeding the Yd11, its internal phase sequence reverses, shifting its output from $+30^\circ$ to -30° . By similarly swapping the output lines, it perfectly mimics a Dy1 transformer and can be paralleled!

A. Connection and Phasor Diagram of Dy1

- **Primary (Δ):** A_2 to B_1 , B_2 to C_1 , C_2 to A_1 .
- **Secondary (Y):** a_1, b_1, c_1 shorted to form neutral n . Outputs from a_2, b_2, c_2 .
- **Phasor:** Secondary line voltage lags primary by 30° (1 o'clock).

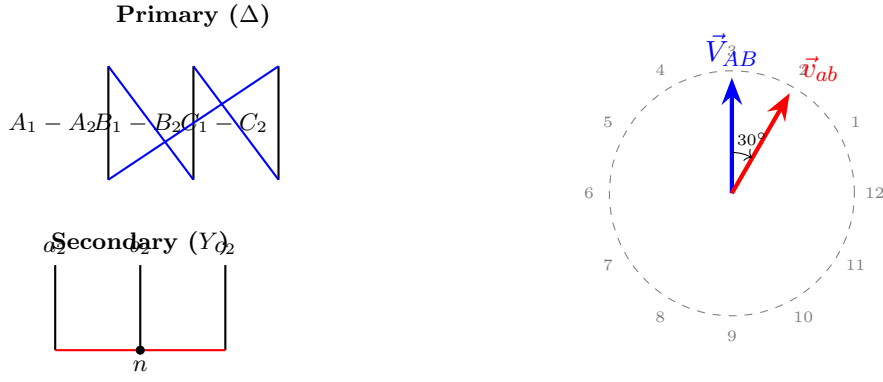


Figure 3: Dy1 Winding and Phasor Diagram.

B. Connection and Phasor Diagram of Yd11

- **Primary (Y):** A_1, B_1, C_1 shorted to form neutral N . Outputs from A_2, B_2, C_2 .
- **Secondary (Δ):** a_1 to b_2 , b_1 to c_2 , c_1 to a_2 .
- **Phasor:** Secondary line voltage leads primary by 30° (11 o'clock).

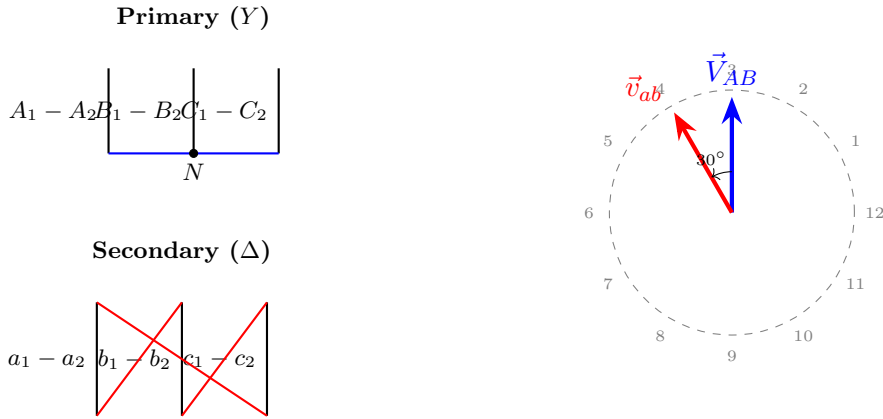


Figure 4: Yd11 Winding and Phasor Diagram.

C. Paralleling Dy1 and Yd11 Successfully Transformers can only be paralleled if they belong to the same vector group phase-shift category (i.e., $\Delta\theta = 0^\circ$). A Dy1 has a -30° shift, and a Yd11 has a $+30^\circ$ shift. Direct connection will result in a 60° phase difference and a massive short circuit.

However, a Yd11 transformer can be **artificially converted into a Dy1 equivalent** by deliberately altering the external phase sequence connections at its terminals.

1. On the primary side of the Yd11 transformer, swap the incoming grid connections for Phase B and Phase C.
2. On the secondary side of the Yd11 transformer, swap the outgoing busbar connections for Phase B and Phase C.

By feeding the transformer with a reversed phase sequence ($A - C - B$ instead of $A - B - C$), the internal magnetic rotation is reversed. This physically reverses the direction of the secondary geometric phase shift. The original $+30^\circ$ advancement (11 o'clock) is inverted into a -30° delay (1 o'clock). Because the output terminals are also swapped back, the Yd11 now acts exactly like a Dy1 transformer at the busbars and can be safely paralleled.

Question 3. Magnetizing Currents & Scott Connection

[7 + 11 = 18 Marks]

Problem Statement 3(i):

Show with a diagram how 3rd harmonics current is produced in a transformer.

Answer:

Thought Process:

1. The transformer core is made of ferromagnetic steel, which exhibits magnetic saturation (non-linear B-H curve).
2. To induce a purely sinusoidal voltage in the windings, the core magnetic flux (ϕ) must be purely sinusoidal.
3. However, due to the flattened, saturating top of the B-H curve, driving a sinusoidal flux to its peak requires a disproportionately large amount of magnetizing force (H , and therefore current i_m) at the peaks.
4. This results in a magnetizing current waveform that is "peaked". Using Fourier analysis, this peaked symmetrical waveform is proven to consist of a fundamental sine wave superimposed with a massive 3rd harmonic current component.

The production of 3rd harmonic current in a transformer is an unavoidable consequence of the ****magnetic saturation**** characteristic of the ferromagnetic core.

For a transformer to function perfectly and induce a clean, sinusoidal secondary voltage without distortion, the mutual magnetic flux $\phi(t)$ in the core must be a pure sine wave:

$$\phi(t) = \Phi_{\max} \sin(\omega t)$$

However, the relationship between the required magnetizing current (i_m) and the resulting flux (ϕ) is dictated by the core's non-linear B-H curve. As the sinusoidal flux approaches its peak value during each half-cycle, the iron core begins to saturate. In the saturation region, creating a small increase in flux requires an exceptionally large increase in magnetizing current.

To force the flux to follow the sinusoidal peak, the magnetizing current must spike sharply. This results in a highly ****peaked, non-sinusoidal current waveform****.

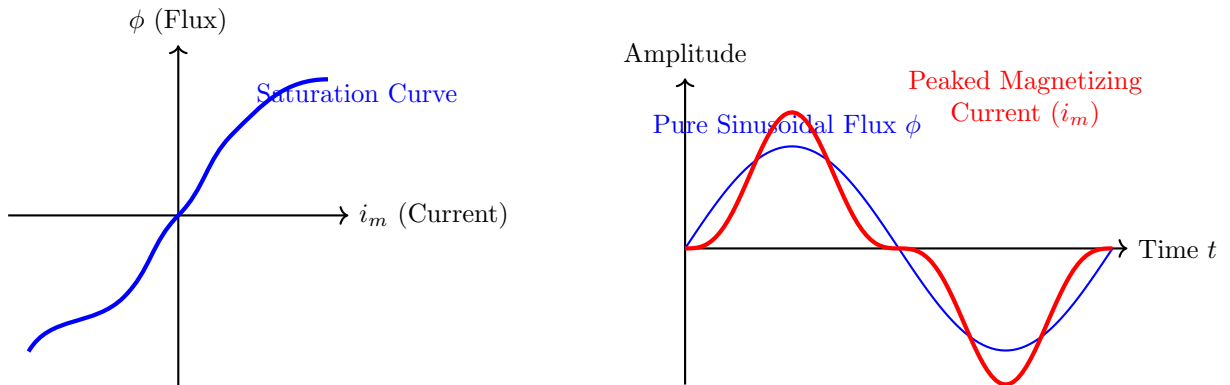


Figure 5: Graphical derivation of the peaked magnetizing current required to produce a sinusoidal flux.

When this peaked, symmetrical current waveform is analyzed using a Fourier series, it resolves into:

$$i_m(t) = I_{1m} \sin(\omega t) - I_{3m} \sin(3\omega t) + I_{5m} \sin(5\omega t) - \dots$$

The most dominant harmonic component is the 3rd harmonic ($I_{3m} \sin(3\omega t)$). Thus, the 3rd harmonic current is internally produced (demanded) by the transformer to satisfy its own magnetic core saturation characteristics.

Problem Statement 3(ii):

In Scott-connected transformers, teaser transformer supplies 0.75 leading power factor load of 25 kW at 220 V and main transformer supplies 0.75 power factor lagging load of 45 kW at 220 V, from a three phase input line voltage of 6600V. Determine the input 3-phase line currents. Draw input current phasors computed for above along with a Scott-connection diagram. Neglect magnetizing currents and the leakage impedance drops.

Answer:

Thought Process:

1. Extract System Constraints:

- Primary line voltage $V_1 = 6600$ V. Secondary voltage $V_2 = 220$ V.
- Base Transformation ratio: $a = N_1/N_2 = 6600/220 = 30$.
- Main primary turns = N_1 . Teaser primary turns = $0.866N_1$.

2. Teaser Output (Phase T):

- $P_T = 25$ kW, $\cos \phi_T = 0.75$ leading $\Rightarrow \phi_T = +41.41^\circ$.
- $I_{2T} = \frac{25000}{220 \times 0.75} = 151.52$ A.
- Vector: $\vec{I}_{2T} = 151.52 \angle 131.41^\circ$ (Relative to Teaser voltage at 90°).

3. Main Output (Phase M):

- $P_M = 45$ kW, $\cos \phi_M = 0.75$ lagging $\Rightarrow \phi_M = -41.41^\circ$.
- $I_{2M} = \frac{45000}{220 \times 0.75} = 272.73$ A.
- Vector: $\vec{I}_{2M} = 272.73 \angle -41.41^\circ$ (Relative to Main voltage at 0°).

4. Primary Line Current Equations (MMF Balance):

- $\vec{I}_A = \frac{N_2}{0.866N_1} \vec{I}_{2T} = \frac{2}{\sqrt{3} \cdot 30} \vec{I}_{2T}$.
- $\vec{I}_B = \frac{N_2}{N_1} \vec{I}_{2M} - 0.5 \vec{I}_A$.
- $\vec{I}_C = -\frac{N_2}{N_1} \vec{I}_{2M} - 0.5 \vec{I}_A$.

Step 1: Calculate Secondary Load Currents The power factor angle is $\theta = \cos^{-1}(0.75) = 41.41^\circ$.

Teaser Transformer (Leading Load):

$$I_{2T} = \frac{P_T}{V_2 \cos \theta_T} = \frac{25000}{220 \times 0.75} = 151.52 \text{ A}$$

Aligning the Teaser phase voltage to the vertical axis (90° baseline), a leading current adds to the angle:

$$\vec{I}_{2T} = 151.52 \angle (90^\circ + 41.41^\circ) = 151.52 \angle 131.41^\circ \text{ A}$$

Main Transformer (Lagging Load):

$$I_{2M} = \frac{P_M}{V_2 \cos \theta_M} = \frac{45000}{220 \times 0.75} = 272.73 \text{ A}$$

Aligning the Main phase voltage to the horizontal axis (0° baseline), a lagging current subtracts from the angle:

$$\vec{I}_{2M} = 272.73 \angle -41.41^\circ \text{ A}$$

Step 2: Map to Primary Line Currents The fundamental turns ratio of the main transformer is $n = N_1/N_2 = 6600/220 = 30$.

1. Line Current A ($\vec{I}_A$): Line A supplies the teaser primary (which is tapped at 86.6% or $\frac{\sqrt{3}}{2}$).

$$\vec{I}_A = \frac{1}{0.866n} \vec{I}_{2T} = \frac{2}{\sqrt{3} \cdot 30} \times 151.52 \angle 131.41^\circ = 5.83 \angle 131.41^\circ \text{ A}$$

Convert to rectangular form:

$$\vec{I}_A = 5.83(\cos 131.41^\circ + j \sin 131.41^\circ) = -3.85 + j4.37 \text{ A}$$

2. Intermediate Main Current Component ($\vec{I}_M$): This is the current reflected purely by the main transformer's load.

$$\vec{I}_M = \frac{1}{n} \vec{I}_{2M} = \frac{1}{30} \times 272.73 \angle -41.41^\circ = 9.09 \angle -41.41^\circ \text{ A}$$

Convert to rectangular form:

$$\vec{I}_M = 9.09(\cos(-41.41^\circ) + j \sin(-41.41^\circ)) = 6.82 - j6.01 \text{ A}$$

3. Line Current B ($\vec{I}_B$): By MMF balance, line B carries the main current plus half of the returning teaser current.

$$\vec{I}_B = \vec{I}_M - 0.5\vec{I}_A = (6.82 - j6.01) - 0.5(-3.85 + j4.37)$$

$$\vec{I}_B = (6.82 + 1.925) + j(-6.01 - 2.185) = 8.745 - j8.195 \text{ A}$$

Convert to polar form:

$$|\vec{I}_B| = \sqrt{8.745^2 + (-8.195)^2} = \mathbf{11.98 \text{ A}}, \quad \angle \tan^{-1}\left(\frac{-8.195}{8.745}\right) = \mathbf{-43.14^\circ}$$

4. Line Current C ($\vec{I}_C$): Line C carries the inverted main current plus half of the returning teaser current.

$$\vec{I}_C = -\vec{I}_M - 0.5\vec{I}_A = (-6.82 + j6.01) - (-1.925 + j2.185)$$

$$\vec{I}_C = -4.895 + j3.825 \text{ A}$$

Convert to polar form:

$$|\vec{I}_C| = \sqrt{(-4.895)^2 + 3.825^2} = \mathbf{6.21 \text{ A}}, \quad \angle 180^\circ - \tan^{-1}\left(\frac{3.825}{4.895}\right) = \mathbf{142.0^\circ}$$

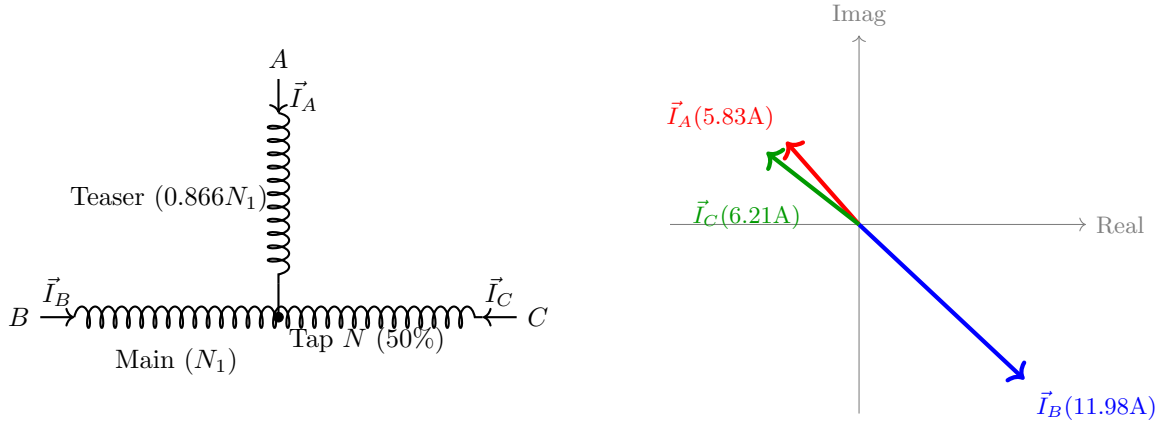


Figure 6: Scott Connection primary winding topology and the resulting unbalanced primary line current phasors.

Question 4. Tap Changers

[7 + 5 + 4 = 16 Marks]

Problem Statement 4(i):

Draw and explain a reactor type on-load tap changer scheme.

Answer:

Thought Process:

1. Reactor type OLTC uses a center-tapped preventive autotransformer (reactor) to transition between taps.
2. Mechanism: Two selector switches (S_1, S_2) ride along the tap studs. They are connected to opposite ends of the reactor. The load is drawn from the reactor's center tap.
3. Explanation: During transition, S_1 and S_2 temporarily bridge two adjacent taps. The reactor's high inductive reactance chokes the resulting circulating fault current, allowing power delivery to remain uninterrupted.
4. I will construct a clean, textbook-accurate TikZ schematic of the mechanism in its bridging state.

The **Reactor-Type On-Load Tap Changer (OLTC)** utilizes a highly inductive center-tapped choke (called a preventive autotransformer or reactor) to safely bridge two adjacent taps during transition without causing a dead short-circuit or interrupting the power supply.

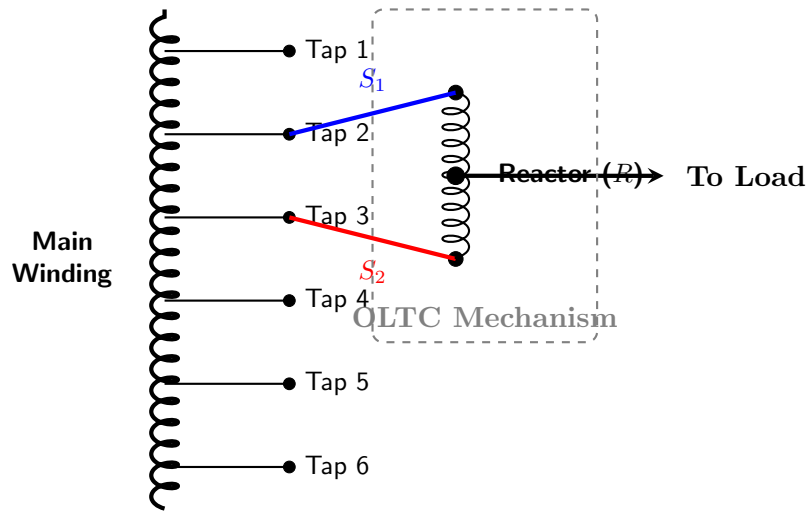


Figure 7: Schematic of a Reactor-Type OLTC showing the transition (bridging) position between Tap 2 and Tap 3.

Operating Sequence (Transitioning from Tap 2 to Tap 3):

1. **Normal Operation (Position 1):** Both selector switches S_1 and S_2 are closed on the same tap (Tap 2). The load current splits equally through the two halves of the center-tapped reactor. Since the currents flow in opposite directions, their magnetic fluxes cancel out, rendering the reactor non-inductive and eliminating voltage drop.
2. **Transition / Bridging (Position 2):** Switch S_2 breaks from Tap 2 and rotates to close on Tap 3. S_1 remains on Tap 2. The reactor is now bridging the voltage difference between Tap 2 and Tap 3. Its high inductive reactance severely chokes the circulating short-circuit current, all while extracting current from the center tap to feed the load without interruption. (As shown in the diagram above).
3. **Completion (Position 3):** Switch S_1 now breaks from Tap 2 and rotates downward to join S_2 on Tap 3. The bridging is complete, fluxes cancel again, and normal operation resumes on the new tap.

Problem Statement 4(ii):

Write the merits and demerits of resistor-type on-load type tap-changers.

Answer:

Thought Process:

1. Resistor OLTCs are the modern standard globally. They are compared against the older Reactor OLTCs.
2. Merits: Resistors transition very fast, limiting arc duration. They operate at unity power factor bridging, which makes the arc easier to extinguish. They are highly compact.
3. Demerits: Because resistors cannot dissipate heat continuously, they will burn out rapidly if the switch mechanism fails and stalls in the bridging position.

Resistor-type OLTCs (like the Flag and Pennant cycles) have largely replaced older reactor-type units in modern high-voltage transformers. Their evaluation is as follows:

Merits (Advantages):

- **Unity Power Factor Arcing:** Transition resistors limit the circulating current purely resistively. Breaking a unity power factor arc is significantly easier and safer than breaking a highly inductive reactive arc, causing far less contact wear.
- **High-Speed Transition:** The diverter switch mechanisms use spring-loaded snap actions, transferring the current in milliseconds. This minimizes the time the arc exists.
- **Compact and Lightweight:** Transition resistors are significantly smaller, lighter, and cheaper to manufacture than bulky iron-core transition reactors.

Demerits (Disadvantages):

- **Thermal Vulnerability:** Resistors are only rated for short-time duty (milliseconds). If the mechanical drive system fails or jams during the transition phase, the continuous bridging current will cause the resistors to overheat and melt down almost instantaneously.
- **No Intermediate Taps:** Unlike reactor systems, the switch cannot be left permanently in the bridging position to provide an intermediate voltage step.
- **Mechanical Complexity:** The requirement for a separate, high-speed, spring-loaded diverter mechanism adds mechanical complexity and requires rigorous routine maintenance.

Problem Statement 4(iii):

Why high voltage winding of a transformer is preferred for tappings of tap-changer?

Answer:**Thought Process:**

1. Why put the tap on the HV side instead of the LV side?
2. Current: $S = \sqrt{3}VI$. HV side has a much lower current. Lower current = smaller OLTC contacts, less arcing, less thermal stress.
3. Construction: In a standard core design, LV is wound next to the core, HV is wound on the outside. It is physically much easier to pull tap leads off the outer layer.
4. Resolution: HV has many more turns. A 1-turn jump on HV is a tiny percentage of total voltage, allowing for fine-grained steps (e.g., 1.25% per step).

Locating the tap changer mechanisms and tap leads on the High Voltage (HV) winding is standard engineering practice for three major reasons:

1. **Lower Current Limits Arcing:** For a given transformer power rating (S), the current on the HV winding is much lower than on the LV winding ($I_{HV} \ll I_{LV}$). A lower nominal current reduces the severity of the arcing across the OLTC contacts during transition. This minimizes contact erosion, limits oil contamination, and allows the manufacturer to use smaller, more cost-effective switchgear.

2. **Structural Accessibility:** In standard concentric winding construction, the LV winding is placed closest to the iron core to minimize the bulk of the required ground insulation. The HV winding is then wound concentrically over the outside of the LV winding. Being the outermost layer, the HV winding is physically accessible, making it easy to route the tap connecting leads out to the tap changer enclosure.
3. **Finer Voltage Regulation:** The HV winding naturally possesses a significantly larger total number of turns ($N_{HV} \gg N_{LV}$). Having more turns allows the manufacturer to offer a higher number of tap steps. Tapping a single turn on the HV side represents a very small percentage change in the voltage ratio, granting the system much finer, higher-resolution voltage control.

Question 5. Short Notes

[8 × 2 = 16 Marks]

Problem Statement 5(i):

Full wave impulse tests on a transformer.

Answer:

Thought Process:

1. What is an impulse test? A high-voltage test designed to simulate a lightning strike hitting the transmission line.
2. Standard waveform: 1.2/50 μ s full wave. Rises to peak in 1.2 μ s, decays to 50% in 50 μ s.
3. Purpose: Verifies the inter-turn insulation and overall dielectric strength of the windings against steep, high-frequency transients.
4. Detection: Done by comparing oscillograms of the applied voltage and the resulting neutral current against reference waves. Any divergence indicates an internal flashover/insulation puncture.

Full Wave Impulse Testing is a critical dielectric type-test performed on power transformers to verify their insulation integrity against severe overvoltages, specifically simulating the effects of lightning strikes on the connected transmission lines.

1. The Standard Impulse Waveform: The test utilizes a standardized artificial lightning wave, defined internationally as a ****1.2/50 μ s full wave****.

- ****Front Time (1.2 μ s):**** The voltage surges extremely rapidly, reaching its absolute peak (crest voltage) in just 1.2 microseconds. This steep front tests the inter-turn capacitance and insulation.
- ****Tail Time (50 μ s):**** The wave then decays exponentially, dropping to exactly 50% of its peak value at 50 microseconds.

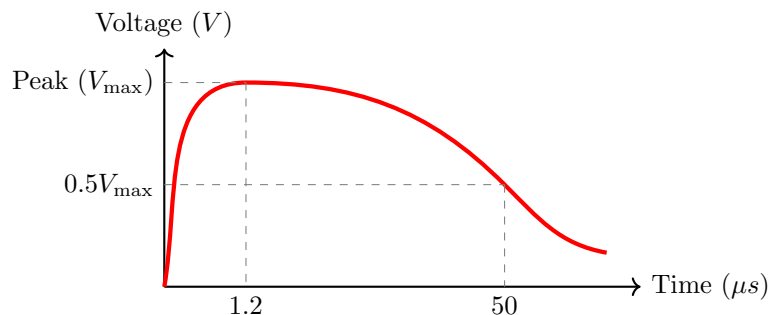


Figure 8: Standard 1.2/50 μ s full wave lightning impulse profile.

2. Testing Procedure and Fault Detection: The impulse voltage is applied to one line terminal of the transformer while the other terminals and the tank are grounded. Because the wavefront is so

steep, it acts as a high-frequency transient (equivalent to ~ 1 MHz). At this frequency, the initial voltage distribution is entirely dictated by the internal winding capacitance, placing immense stress on the first few turns of the coil.

To determine if the transformer passes, engineers record high-speed oscillograms (digital traces) of both the applied voltage and the current flowing out of the neutral terminal to ground. If the insulation punctures or a flashover occurs between turns, the internal impedance drops drastically. This causes clear distortions, ripples, or sudden drops in the neutral current trace when compared to a low-voltage reference trace, signaling failure.

Problem Statement 5(ii):

Neutral oscillation in 3-phase transformers & Tertiary windings.

Answer:

Thought Process:

1. Neutral Oscillation (Inversion): Occurs in ungrounded Star-Star systems during severe line-to-ground faults or heavy single-phase unbalances.
2. Mechanics: The neutral point is electrically "floating". If phase A shorts to ground, the neutral point N shifts forcibly to the potential of line A. This distorts the voltage triangle, heavily overvoltageing the healthy phases B and C (up to $\sqrt{3}V_{ph}$).
3. Tertiary Winding Solution: Adding a closed Delta (Δ) tertiary winding provides a circulating path for zero-sequence currents.
4. By allowing zero-sequence currents to flow, the tertiary winding acts as an electromagnetic anchor, pinning the neutral point securely to the geometric center of the phasor triangle and preventing hazardous oscillations.

1. Neutral Oscillation (Neutral Inversion): In three-phase Star-Star ($Y - Y$) transformers where the primary neutral is ungrounded (floating), the electrical neutral point is highly unstable. Under normal, perfectly balanced conditions, the neutral rests at the exact geometric center of the voltage triangle, maintaining a potential of zero volts.

However, if a severe asymmetrical fault occurs—such as a single line-to-ground fault on Phase A—or a massive single-phase load is applied, severe zero-sequence voltages are generated. Because the neutral is ungrounded, the zero-sequence currents cannot flow to counterbalance this. Consequently, the neutral point (N) physically "shifts" or oscillates away from the center of the phasor diagram toward the faulted phase. If Phase A hits ground potential, N shifts to A . As N moves, the phase voltages across the healthy windings (V_{BN} and V_{CN}) expand drastically, jumping from their nominal value to a full line-to-line value ($\sqrt{3}V_{ph}$). This severe, oscillating overvoltage can instantly destroy the insulation on the healthy phases.

2. The Role of the Tertiary Winding: The standard engineering solution to suppress neutral oscillation is the addition of a **Tertiary Winding** connected in a closed **Delta (Δ)** configuration.

When an unbalance attempts to shift the neutral point, it generates zero-sequence currents. The closed delta loop of the tertiary winding provides a localized, low-impedance path that allows these zero-sequence currents to circulate freely inside the transformer.

By allowing these balancing currents to flow, the tertiary winding magnetically counteracts the unbalance. It acts as a rigid electromagnetic anchor, locking the floating neutral point (N) securely to the geometric center of the voltage triangle, entirely eliminating neutral oscillation and protecting the system from asymmetric overvoltages.

— END OF SOLUTIONS —